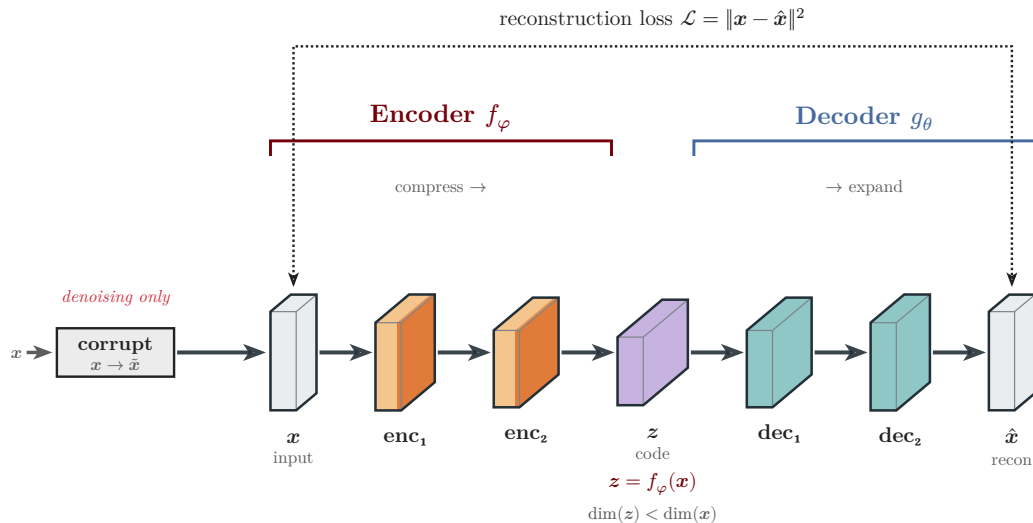




The four family members share one scaffold — encoder $\rightarrow$ code $\rightarrow$ decoder — and differ in the colored add-on

Undercomplete

bottleneck dim < input dim;
compression is the only regularizer

objective:
 $\|\mathbf{x} - \hat{\mathbf{x}}\|^2$

Denoising

corrupt input $\mathbf{x} \rightarrow \tilde{\mathbf{x}}$,
reconstruct the clean $\mathbf{x}$

objective:
 $\|\mathbf{x} - g_\theta(f_\varphi(\tilde{\mathbf{x}}))\|^2$

Sparse

penalize code activations $\Omega(\mathbf{z})$
so few units fire per input

objective:
 $\|\mathbf{x} - \hat{\mathbf{x}}\|^2 + \lambda\Omega(\mathbf{z})$

Masked

hide a subset of input
tokens / patches, predict them

objective:
 $\sum_{i \in \mathcal{M}} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2$